MIL-C-2121A(MU)
1 NOVEMBER 1963

SUPERSEDING
MIL-C-2121
29 MARCH 1950
FA(FC)-PD-159
9 MAY 1961

MILITARY SPECIFICATION

CASE, FIRE DIRECTION SET, TANK AND TANK DESTROYER WEAPONS

1. SCOPE

1.1 This specification covers one type of carrying case to contain all items used with the fire direction set for tank companies.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-F-106 —Fasteners, Slide, Interlocking.
V-T-276 —Threads, Cotton.
CCC-C-419 —Cloth, Cotton, Duck, Unbleached, Plied Yarns (Army and Numbered).

MILITARY

MIL-C-496 —Clips, End.
MIL-W-530 —Webbing, Textile Cotton, General Purpose Natural or in Colors.
MIL-B-543 —Buckles, Tongueless, and Web Strap.
MIL-F-10884 —Fasteners, Snap.
MIL-F-13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.
MIL-P-14232 —Parts, Equipment and Tools for Ordnance Materiel, Packaging of.
MIL-I-45208 —Inspection Requirements, General Specification for.

STANDARDS

FEDERAL

FED-STD-595—Colors.
FED-STD-751—Stitches; Seams; and Stitching.

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

FSC 1290

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:24Z
Check the source to verify that this is the current version before use.

MIL-STD-130 —Identification Marking of U. S. Military Property.

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

A8636242 —Carrying Case, Fire Direction Set, Tank and Tank Destroyer Companies, Canvas.

PACKAGING DATA SHEET

MIL-P-14232/P8636242 —Case, Fire Direction Set, for Tank Companies.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.1.1 *Cloth, cotton, duck.* Cloth duck numbers 6 and 10 shall be of hard texture conforming to Type 1, Table I of Specification CCC-C-419.

3.1.1.1 *Color and treatment.* Cloth duck, webbing, thread, and all tape (including slide fastener tape) shall be water repellent, vat-dyed olive drab and matched to the color chip in Standard FED-STD-595 under Table VI, No. 34102; and shall be fungi-static in accordance with the fungus resistance requirements of Specification MIL-F-13926.

3.1.2 *Webbing, cotton.* Webbing material of case components shall conform to the following requirements of Specification MIL-W-530:

(a) Type I, ¾ inch width — for thongs, seam reinforcements on gusset, and all binding except outside edges of case.
(b) Type I, 1 inch width — for tabs and binding outside edges of case.
(c) Type I, 1½ inch width — for dust flaps.
(d) Type III, ⅝ inch width—for billets "B" and "C", buckle chapes "B" and "C", and binding straps.
(e) Type III, 1 inch width — for buckle chapes "A", billet "A", handles, and handle reinforcements.

3.1.3 *Thread, cotton, machine.* Machine cotton threads, ticket number 10/3, 3 ply stitch conforming to Type IIIA, Table V and Type IIIB, Table VI of Specification V-T-276.

3.1.4 *Fasteners, snap.* Fasteners shall be brass and made in accordance with Style 2, Finish 2 of Specification MIL-F-10884.

3.1.5 *Clips, end.* Clips shall be brass with a chemical black finish and made in accordance with Type I of Specification MIL-C-496.

3.1.6 *Buckles, tongueless, 2 bar.* Buckles shall be of ⅝ inch and 1 inch sizes conforming to Type II, Class I, Style 4, Construction B and Type II, Class I, Style 1 respectively of Specification MIL-B-543.

3.1.7 *Fastener, slide.* Fasteners shall be in accordance with Type I, Style 18, two nonreversible single pull sliders, Class A, heavy size, wire stirrup pull, one inch tape extension, dark oxidized brass metallic parts of Specification V-F-106.

3.2 Fabrication. The case shall be manufactured in accordance with Drawing A8636242, and drawings pertaining there-

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:24Z
Check the source to verify that this is the current version before use.

to. Requirements of this specification are detailed only to the extent considered necessary to insure inherent quality and performance capabilities.

3.2.1 *Stitching.* All stitching shall be machined, Type 301, 8 stitches to the inch, in accordance with Standard FED–STD–751.

3.2.1.1 *Thread type.* Type IIIB thread shall be used for needle or top thread and Type IIIA for bottom or bobbin thread in machine.

3.2.1.2 *Machine stitching.* When machine stitching is not continuous the thread shall be backstitched not less than 1 inch at each break. All stitching shall be backstitched at the ends to prevent ravelling, except when ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained so that the lock will be embedded in the center of the materials sewed.

3.3 Identification and marking.

3.3.1 *Identification.* Each pocket shall be identified by specific numerical designation and located in accordance with requirements of applicable drawings. "U.S." markings shall be located accordingly (see 3.2).

3.3.2 *List of contents.* The following list of contents shall be identically printed within the area provided on pocket "16" of appropriate drawing:

Pocket	*Contents*	*Quantity*
1	Thumbtacks, ½", 100 to a box	1 box
2	Erasers, rubber, art gum	6 ea
3	Maptacks, colored spherical head:	
	Blue	1 box
	Green	1 box
	Red	1 box
	White	1 box
4	Erasers, rubber, pencil, oblong, beveled.	6 ea
5	Tape, cellulosic, transparent, ¾" x 36 yd.	1 roll
6	Scale, plotting, triangular, 12"	1 ea
7	Forms, record, 100 sheets to pad	2 pads
8	Forms, data sheet, 100 sheets to pad.	1 pad
9	Rule, slide, w/case	1 ea
10	Pencils, glazed surface marking:	
	Blue	6 ea
	Red	6 ea
	Pencils, wood cased, colored:	
	Blue	2 ea
	Brown	2 ea
	Green	2 ea
	Red	2 ea
	Yellow	2 ea
11	Pencils, glazed surface marking:	
	Black	6 ea
	Pencils, wood cased, drawing:	
	3H	6 ea
	7H	6 ea
12	Straight edge, 30"	1 ea
13	Scale, coordinate	1 ea
14		
15	Maptacks, rectangular head:	
	¾" OA length	1 pkg
	1⅛" OA length	1 pkg
16	Protractor, semicircular	1 ea
17	Protractor, fan	2 ea
Front	Plotting sheets, gridded, 1/25000 scale:	
	22" x 30", paper, planetable	4 doz
	22½" x 30", plastic, coated	8 ea
Back	Paper, tracing, white, 18" x 24"	100 shts

3.3.3 *Markings.* Individual pocket numbers referenced in 3.3.1 shall be ¼ inch in size. All tabular markings on pocket "16" shall be 5/32 inch except the word "contents" shall be ¼ inch. Other identification and marking shall be in accordance with Standard MIL–STD–130. Markings to be of black waterproof ink or paint.

3.4 Environmental .

3.4.1 *Storage temperatures.* The case shall show no evidence of physical failure and shall meet requirements of 3.7 and 3.8 at room temperature (plus 60 degrees to plus 90 degres Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:24Z
Check the source to verify that this is the current version before use.

3.4.2 *Rain.* The case shall be capable of withstanding a rainfall of 4 inches per hour for a period of 2 hours, without visible damage or defect, and shall subsequently meet the following requirements of this specification.

3.5 Fasteners, snap. Fasteners shall be clinched tight. Punched holes shall provide a snug fit for insertion of the fastener tubes.

3.6 Clips, end. Clips shall be securely attached without cutting the webbing. The webbing shall be inserted the full depth of the clip and be flat at the point of entrance.

3.7 Fastener, slide. The thongs shall permit easy operation of the fastener to open and close the case, and there shall be no restriction or binding throughout the 7 foot chain.

3.8 Handles, carrying. The carrying handles shall be convenient to grip, securely attached, and sufficient in strength to support at least 300 pounds when carried.

3.9 Workmanship. The finished case shall be clean, free from any defect which may affect appearance or serviceability, and shall be representative of quality workmanship and application of appropriate manufacturing processes in fulfillment of requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. Except where otherwise specified herein, the contractor's inspection system shall be in accordance with Specification MIL–I–45208.

4.3 Lot information. When inspection lot sizes and lot formations are applicable, they shall be in accordance with Standard MIL–STD–105 and shall also be defined in the contractor's inspection system.

4.3.1 *Defective units and lots.* Defective units and lots shall be processed as specified in Standard MIL–STD–105 and shall not be included as a part of the quantity specified in the contract until formal approval is obtained.

4.4 Inspection provisions. Components and sub-assemblies shall be examined in accordance with the inspection provisions contained in Specification MIL–F–13926 and 4.4.1 herein. Test and control of materials not covered by Supplementary Quality Assurance Provisions (SQAP's) or Section 4 of Specification MIL–F–13926 shall be in accordance with "Test Data" provisions of Specification MIL–I–45208. Inspection related to Section 3 of this specification shall be performed in accordance with 4.4.2. The tabulated Classification of Defects listed herein shall constitute the minimum inspection to be performed by the contractor prior to Government acceptance. The Government reserves the right to inspect for any applicable requirements, and to reject individual non-conforming items.

4.4.1 *Inspection of components.* All positional and functional characteristics shown on drawings, or contained in notes on the drawings or referenced specifications, which indicate precision requirements and have a bearing on the functional accuracy and performance of the Case shall be inspected in accordance with the written inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:24Z
Check the source to verify that this is the current version before use.

system required by Specification MIL–I–45208. These characteristics are generally defined in Specification MIL–F–13926 as Class 1 or Class 2 Defects.

4.4.1.1 *Component sampling inspection.* Where quality control has been established on in-process manufacture, sampling inspection of components may be used in lieu of 100 percent inspection and shall be in accordance with the sampling plan tables specified in Specification MIL–F–13926.

4.4.2 *Classification of defects and sampling plans.* Examination and testing to determine acceptance or rejection in compliance with the applicable requirements of Section 3 shall be conducted on a 100 percent inspection basis, initially. After conclusive evidence of the process average has been established in conformance with Standard MIL–STD–105 and the AQL's specified herein, the classification of defects and sampling plans specified in Table 1 may be used in lieu of 100 percent inspection. The AQL value specified shall apply to each individual characteristic on a single defect basis.

4.4.2.1 *Disposition of rejected lots.* Rejected lots shall be screened in conformance with 4.3.1 for all defective characteristics causing rejection of the lot. After the Government inspector has approved the corrective action taken, subsequent production lots may be submitted for acceptance in accordance with 4.4.2 and Standard MIL–STD–105.

TABLE I. *Classification of defects*

Use Inspection Level II in Table III with Sampling Plan Table IV–A of Standard MIL–STD–105.

Critical: None defined.

Major: AQL 1.0 percent defective.	*Req't par.*	*QA par.*	*Test method*
101. Improper identification and markings	3.3	4.6.2.2	Std meas equip (SME)–Visual
102. Snap fastener failure	3.5	4.6.2.5	Visual–Tactile
103. End clips insecure	3.6	4.6.2.6	Visual–Tactile
104. Slide fastener inoperative	3.7	4.6.2.7	Tactile
105. Carrying handle insecure	3.8	4.6.2.8	Visual–Tactile
106. Poor workmanship	3.9	4.6.2.9	Visual–Tactile

Minor: None defined.

Note. The above tests shall be conducted at standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit) in accordance with the quality assurance paragraph (QA par.) listed.

4.4.3 *Control inspection.* The first 2 cases produced, and 3 selected at random from every 100 subsequently produced (or from each month's production, whichever occurs first) shall represent the control samples for control inspection. Control samples shall be subjected to the examinations and tests specified herein to determine compliance with 3.1, 3.2, and 3.4. The examinations and tests shall be conducted by the contractor under surveillance of the Government representative at the contractor's facility, or Government approved laboratory, using test equipment authorized for use by the contract, or otherwise approved by the contracting officer.

4.4.3.1 *Control sample failure.* Should a control sample fail to meet the requirements specified in 4.4.3, the contractor shall cease those operations causing rejection until necessary corrections have been made by the contractor and approved by the Government. When production has resumed, inspection of those requirements that have previously caused rejection shall be performed on a 100 percent basis until 5 consecutively produced cases meet these

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:24Z
Check the source to verify that this is the current version before use.

requirements. When this is accomplished, control inspection (4.4.3) may be continued.

4.5 Inspection equipment. Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL–I–45208.

4.5.1 *Accuracy requirements.* Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of permissible tolerances except as otherwise specified. In cases where ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6 Test methods and procedures.

4.6.1 *Materials.* The materials shall be inspected in accordance with the quality assurance provisions of the applicable specifications to determine compliance with 3.1 and 4.4.3.

4.6.2 *Fabrication.* The case shall be subjected to a visual, tactile and dimensional inspection to determine conformance to 3.2 and the applicable drawings.

4.6.2.1 *Stitching.* The stitching of the case shall be inspected in conformance with the procedures specified in Federal Standard FED–STD–751 in compliance with 3.2.1 for the specified type machine thread and quality of workmanship.

4.6.2.2 *Identification and markings.* Markings shall be visually inspected for legibility and accuracy. Size and position of the markings shall be measured with standard measuring equipment. Identification and markings shall meet the requirements specified in 3.3.

4.6.2.3 *Storage temperatures.* The case shall be tested in a chamber capable of maintaining the temperature requirements of 3.4.1. The test shall be conducted in accordance with Procedure II of Specification MIL–F–13926. After exposure, the case shall be visually inspected for damage or physical failure in compliance with 3.4.1. Subsequent to this test the case shall be subjected to the tests specified in 4.6.2.7 and 4.6.2.8 and shall meet the requirements specified therein.

4.6.2.4 *Rain.* The rain test shall be conducted in accordance with Procedure I of Specification MIL–F–13926. Subsequent to the rain test, the case shall be subjected to a visual examination and shall meet the requirements specified in 3.4.2. The case shall then be subjected to the tests specified in 4.6.2.5 to 4.6.2.8 inclusive and shall meet the requirements specified therein.

4.6.2.5 *Fasteners, snap.* The fasteners shall be subjected to a visual and tactile examination. The male and female half shall be correctly assembled and tightly clinched to the material in compliance with the requirements of 3.5. The fastener halves shall be manually snapped together and unsnapped at the point normally used when opening and closing. The fastener shall be manually engaged and disengaged for not less than 10 times to determine tightness and snugness of fit in compliance with 3.5.

4.6.2.6 *Clips, end.* Clips shall be subjected to a visual and tactile examination for conformance with 3.6. Clips shall be securely attached at the specified distance. A manual pull shall be applied to the end clip and webbing by gripping the webbing with one hand and the clip end with the fingers of the other hand. Any movement of the clip on the webbing shall be cause for rejection.

4.6.2.7 *Fastener, slide.* The slide fastener

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:24Z
Check the source to verify that this is the current version before use.

shall be subjected to a tactile test. The case shall be opened and closed a minimum of 3 times by drawing the slider back and forth by hand over the entire length of the chain. There shall be no interference throughout the entire length of the chain in compliance with 3.7.

4.6.2.8 *Handles, carrying.* The carrying handles shall be inspected by a visual and tactile examination initially to determine compliance with 3.8. Weight equivalent to 300 pounds shall be placed inside the finished case. The case shall then be suspended by the handles and subjected to a continuous pendulum motion for fifty complete cycles at an amplitude of at least one foot (two feet total excursion). Subsequent to this test, the handles shall be examined to determine any detrimental effects or indication of poor workmanship.

4.6.2.9 *Workmanship.* The case shall be given an overall visual and tactile examination for conformance with the requirements specified in 3.9.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8636242. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The case is primarily a protective case intended to house and carry fire direction equipment and accessories used in the operation of a fire direction center for tank companies.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of preservation, packaging and packing.

(c) Applicable stock number.

6.3 Design information. Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

Custodian:

Army—(MU)

Preparing activity:

Army—(MU)

Project No. 1290-A120Y

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:24Z
Check the source to verify that this is the current version before use.